Author

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Greetings!, Im a Dual Degree Student Pursuing BE CSE at a Private College along with this Degree. I'm a person of passion and dedication. exploring and pushing the bounds of creativity, open to implementing daily learnings and reflecting on constructive criticisms professionally. Known for a positive attitude and a sharp sense of humor, I'm confident, trustworthy, and stand on morals. Aspiring to create innovative apps utilizing UI/UX design, AIML, and explore AR/VR innovations, while expressing my creativity through music, storytelling, and design. My Career Goal is to build transformative experiences via tech at the intersection of creativity and innovation.

AI/LLM usage

I used ChatGPT to clarify Flask concepts, SQLAlchemy relationships, validations, and general debugging approaches.

The AI mainly helped with architectural guidance and improving the structure of routes and templates.

Approximately 35–40% influence, limited to suggestions and example code patterns.
All final integration, testing, debugging, and implementation were done manually by me.

Description

This project implements a basic Hospital Appointment Management System with three roles: Admin, Doctor, and Patient.
Patients can book or cancel appointments, doctors can update diagnosis/prescription, and admins can manage doctors and view system-wide data.

This project follows the specifications given for MAD-1, including role-based access, CRUD operations, validations, and dashboard summaries.

Technologies used

- **Python (Flask)** – core backend framework
- **Flask-SQLAlchemy** – ORM + SQLite database
- **Flask-Login** – session management & RBAC
- **Flask-Bcrypt** – password hashing
- **SQLite** – lightweight database
- **Jinja2** – templating engine
- **Bootstrap 5** – front-end styling

Purpose:
Flask provides a simple and modular structure ideal for MAD-1.
SQLAlchemy ensures safe database access.
Flask-Login and Bcrypt handle secure authentication.
Bootstrap improves the UI without heavy CSS.

DB Schema Design

User Table

Column	Type	Details
id	Integer	Primary Key
username	String	Unique, required
password	String	Bcrypted hash
name	String	Full name
email	String	Auto-generated based on username
role	Integer	1=Admin, 2=Doctor, 3=Patient
specialization	String	Used only for doctors

Reasoning:
A single `User` table with a role field simplifies authentication and avoids duplicate user structures.

Appointment Table

Column	Type	Details
id	Integer	Primary Key
doctor_id	Integer	Foreign Key → User.id
patient_id	Integer	Foreign Key → User.id
date	String	YYYY-MM-DD
time	String	HH:MM
status	String	Booked / Completed / Cancelled
diagnosis	Text	Doctor-filled
prescription	Text	Doctor-filled

Reasoning:
Appointments are linked to both doctor and patient using foreign keys.
Diagnosis/prescription allow storing treatment history as required in the spec.

API Design

Although the project primarily uses Flask routes with templates, API endpoints follow clear responsibilities:

- `/login`, `/registerPatient` → authentication
- `/admin/doctors/*` → CRUD operations
- `/patient/appointments/*` → booking/cancellation
- `/doctor/appointments/<id>` → update diagnosis/prescription

A YAML file is included separately to describe each endpoint (as required).
It covers request methods, inputs, and expected flows.

Architecture and Features

Project Organization

- **`app.py`** contains all controllers (routes), authentication logic, and role-based access control.

- **`dba.py`** defines database models `User` and `Appointment`.

- `templates/` holds all Jinja HTML pages (dashboard, login, doctor CRUD, booking page, list views).
- `static/` holds the Bootstrap-based CSS additions.

Key Features Implemented

- **Role-based dashboards** for Admin, Doctor, Patient
- **Admin**:
    - Add/Edit/Delete doctors
    - View all appointments
    - Dashboard metrics (total doctors, patients, appointments)
- **Patient**:
    - Register/Login
    - Book appointments (with double-booking validation)
    - See available departments (specializations)
    - Cancel appointments
    - View complete history incl. diagnosis/prescription
- **Doctor**:
    - See all assigned appointments
    - Update appointment status
    - Add diagnosis + prescription

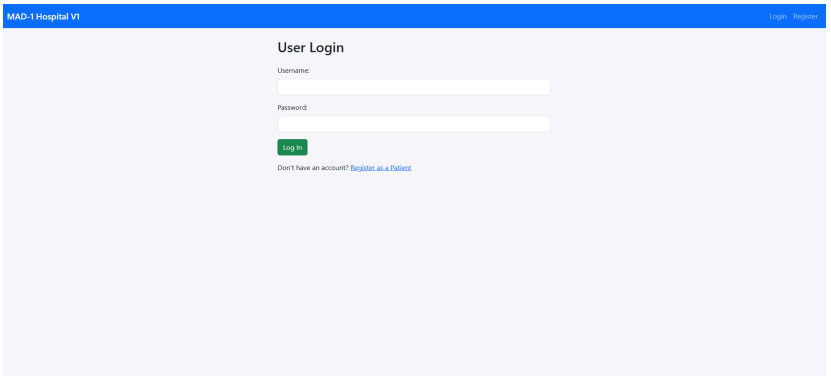
Video

https://drive.google.com/file/d/1Qrme0viLiXtkFS1mtIxRYgZwPdewplOo/view?usp=sharing

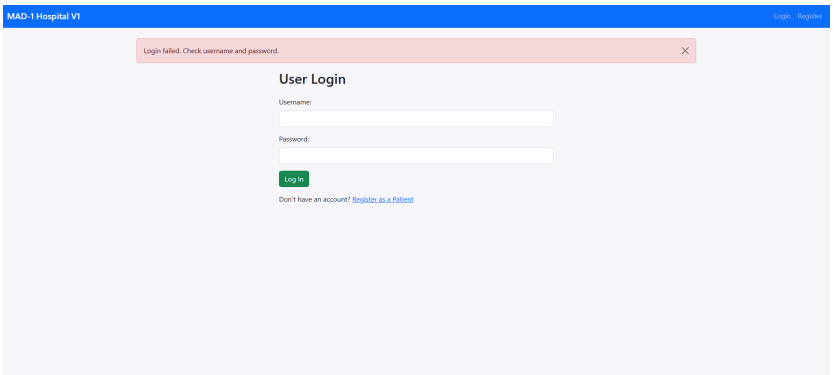
Screenshots

1. Authentication

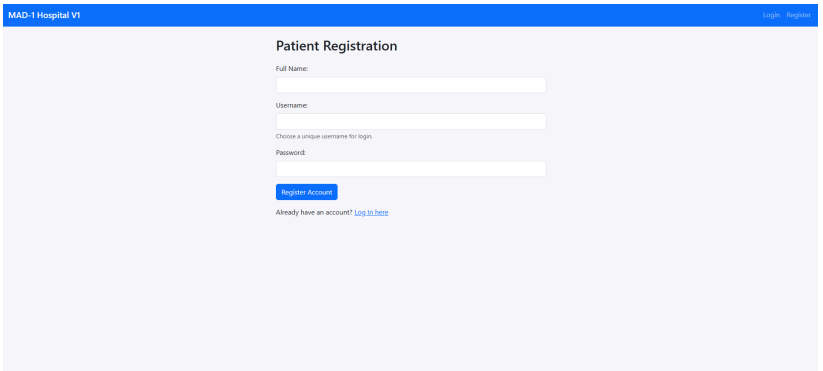
Login Page



Invalid Login Attempt

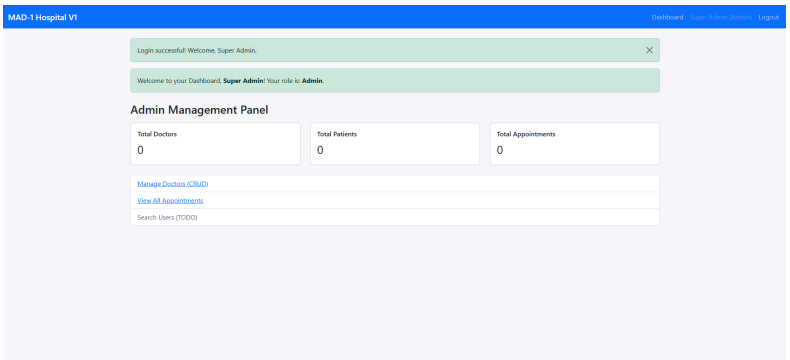


Patient Registration Page

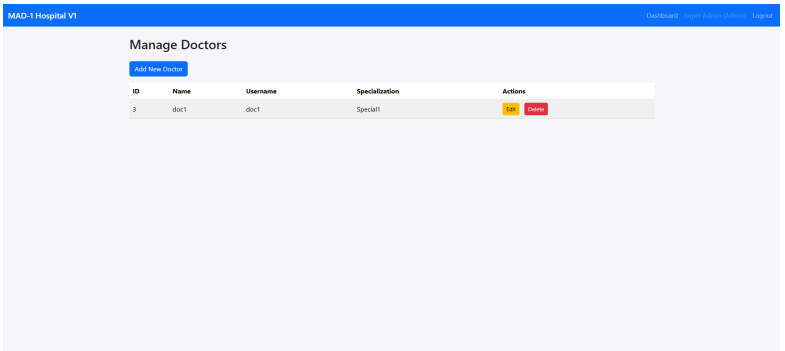


2. Admin Module

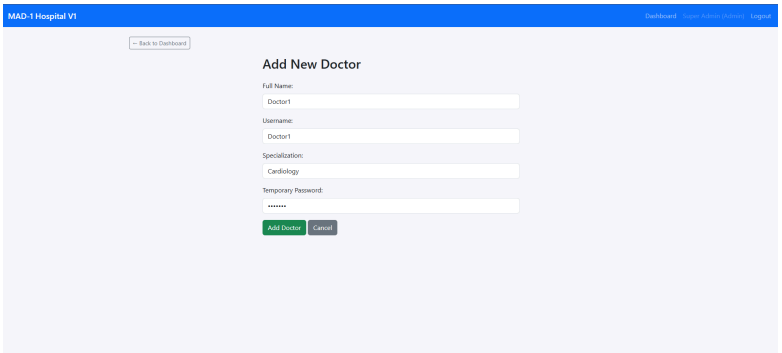
Admin Dashboard (cards showing total doctors/patients/appointments)



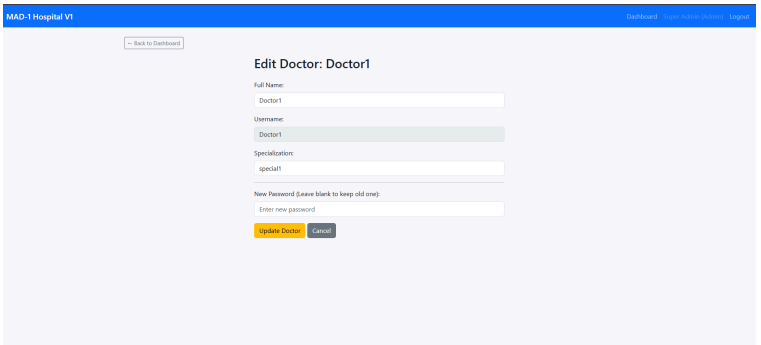
Manage Doctors (CRUD)



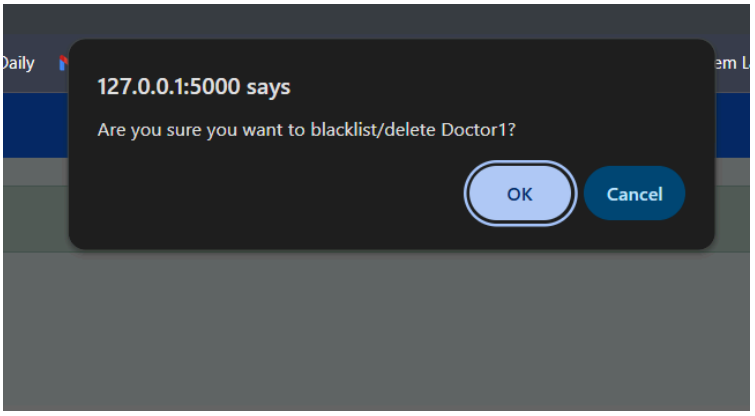
Add Doctor Page



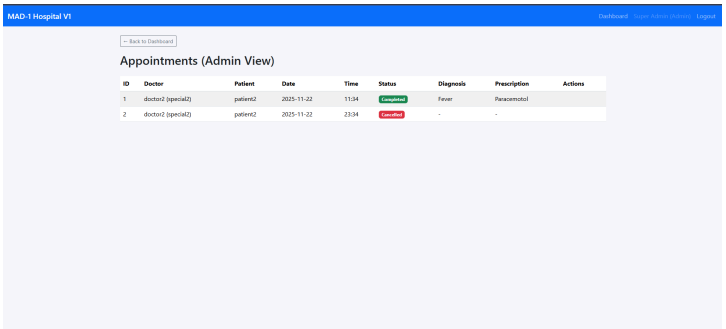
Edit Doctor Page



Delete Doctor Confirmation

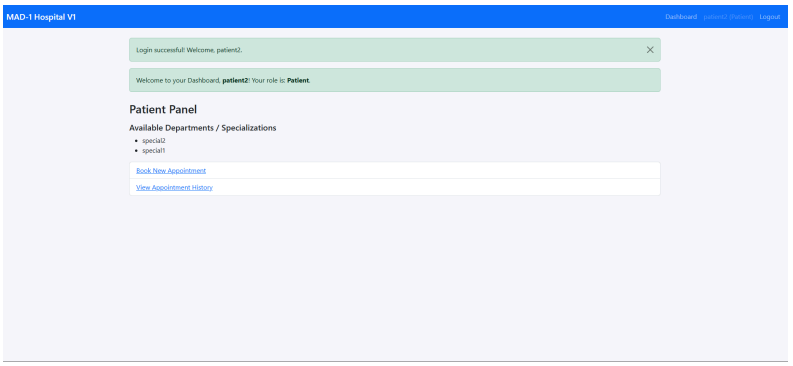


Admin Appointments

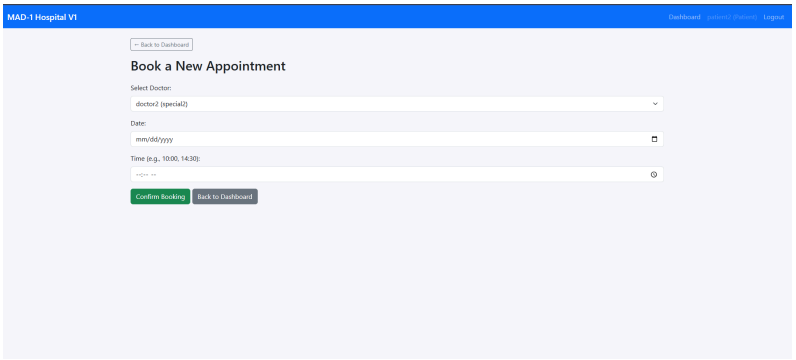


3. Patient Module

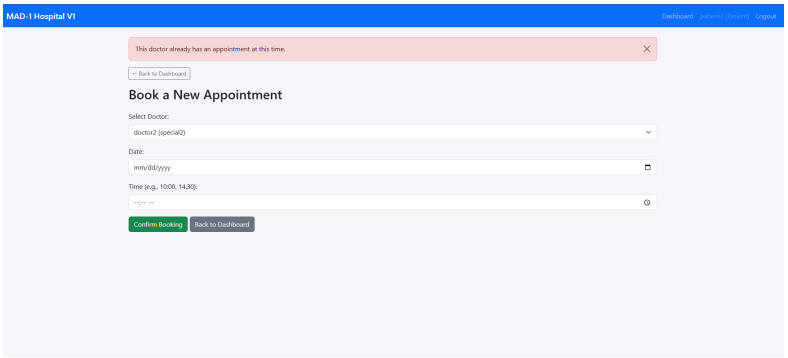
Dashboard



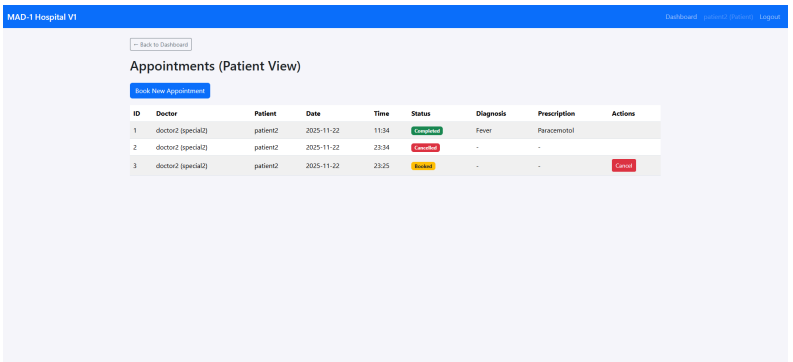
Book Appointment Page



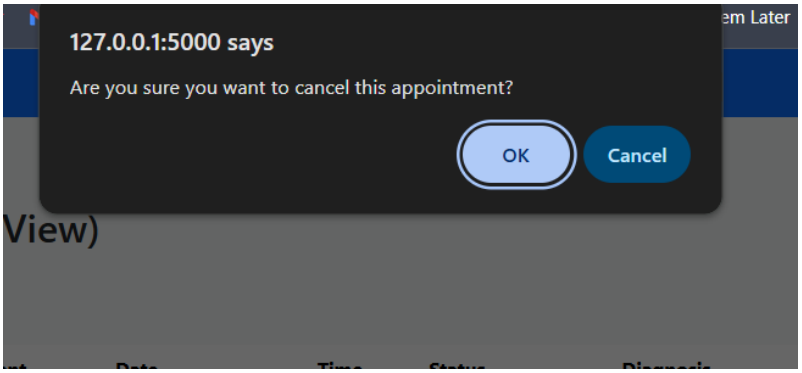
Double-booking Error



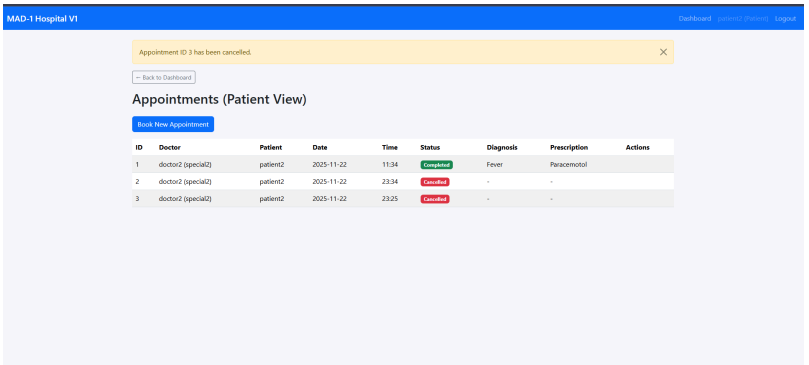
Appointment History



Cancel Appointment Action

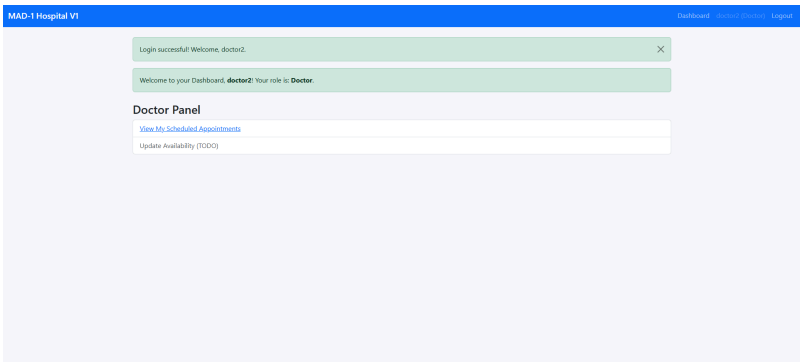


Cancel Success / Updated Status

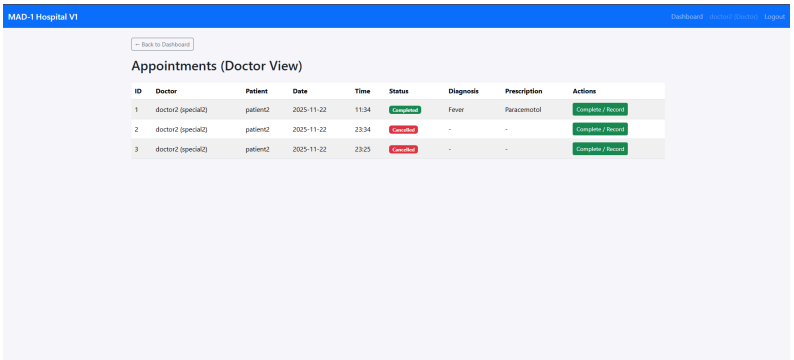


4. Doctor Module

Dashboard



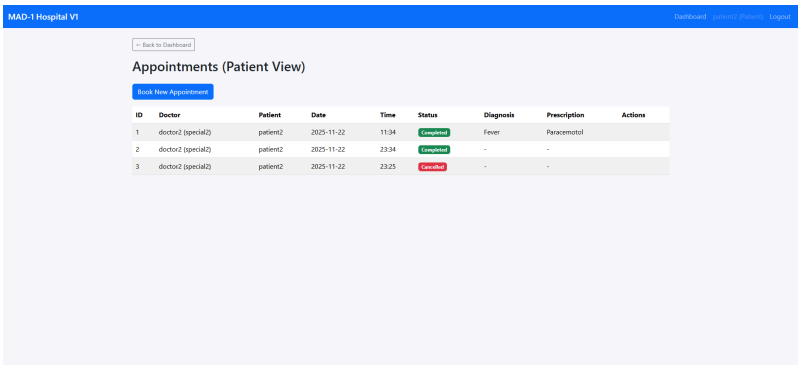
Appointment Handling

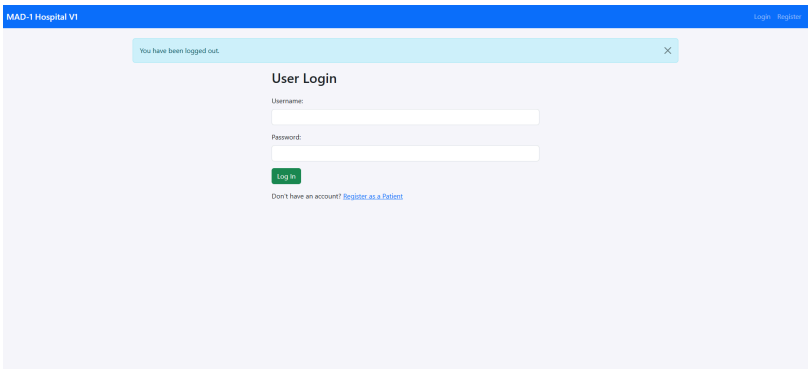


Update Appointment Page (status, diagnosis, prescription)

5. Final Patient View

Patient Appointment History Showing:



6. Logout

Logout / Redirect Page